

Spectral correction in yield forecasts for two-terminal perovskite–silicon tandem modules

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Abstract — The electrical efficiency of two-terminal tandem (2TT) solar cells depends strongly on the incident spectrum. This makes it difficult to forecast the energy yield (EY) of a system containing 2TT modules because the solar spectrum varies continuously, both for direct and diffuse sunlight. The impact of spectrum is often quantified by a spectral correction factor f_λ , which describes how a module's efficiency in the field varies from its efficiency under the AM1.5g spectrum. We use sophisticated EY software to predict f_λ for bifacial perovskite–silicon 2TT modules installed on a single-axis tracker (the most prevalent power-plant configuration installed in 2024) and show that f_λ can deviate from unity by up to 5% for the annual yield, and by up to 20% under some conditions. The implications for cell design are discussed. We also describe how f_λ depends on atmospheric conditions, how 2TTs compare to Si modules, how the spectral dependence decreases once DC/AC inverters are included, and how f_λ can be incorporated into PVSyst. Finally, we calculate the bifacial gain, which is necessarily smaller for 2TT than for Si modules.

I. INTRODUCTION

After a solar module is measured in the factory under standard testing conditions (STC), it is binned, sold, shipped and installed in a field where it will never be subjected to STC again. In the field, the modules operate under temperatures, irradiances, spectra and incident angles that can differ markedly from STC. Thus, to forecast the energy yield (EY) of an installed module, one needs a model that predicts (i) the field conditions and (ii) the module behavior under those conditions; this is the purpose of EY simulation software.

Many studies investigate how two-terminal tandem (2TT) devices would behave once installed in the field [1–11]. They find that the natural variation in incident spectra causes a significant reduction in operating efficiency due to current-mismatch between sub-cells [9]; that the optimal bandgap for the perovskite top sub-cell is larger for real-world conditions than for STC [3, 8]; that the operating temperature has only minor effects on current mismatch [10]; and that the reduction in power due to mismatch is less than the reduction in current due to increases in the fill factor [1, 8].

In this paper, we continue the investigation into 2TTs by simulating their behavior when installed on a single-axis tracker (SAT). We chose a bifacial SAT because it constitutes the most common configuration for modern-day PV power plants, and because, to our knowledge, the behavior of 2TT modules on bifacial SATs has not previously been published.

We quantify how spectral changes alter the behavior of ‘typical’ perovskite–silicon 2TT modules with the spectral correction factor f_λ , defined as

$$P = f_\lambda \times P_{AM1.5g}, \quad (1)$$

where P is the output power and $P_{AM1.5g}$ is the output power if the module were operating under the AM1.5g spectrum. Thus, when f_λ is greater than unity, the efficiency of the module in the field will be higher than its efficiency under the AM1.5g spectrum.

This correction factor is close to unity for c-Si PERC modules: the annual f_λ falls within $\pm 2\%$ [12, 13] and at any time of the day or year, f_λ falls within about ± 5 –10% of unity [12]. By contrast, f_λ for 2TT modules deviates much further from unity because these modules depend more strongly on the incident spectrum [3, 7].

We also compare f_λ of 2TT and PERC modules, examining the trends of f_λ with season, location and atmospheric conditions. We compare the bifacial gain and, like [5], we demonstrate how the gain for 2TTs is much smaller than for PERCs. Finally, we describe how the calculated f_λ can be used in EY programs that do not synthesize their own spectra, like PVSyst.

II. MODELLING APPROACH

Simulations were performed using [SunSolve-Yield](#) [14], which uses ray tracing to solve the optical behavior of the system throughout a year in any location. As described in [15], it accounts for the optical properties, dimensions and configuration of every component within the solar cells, modules and system, solving the annual yield in one-hourly

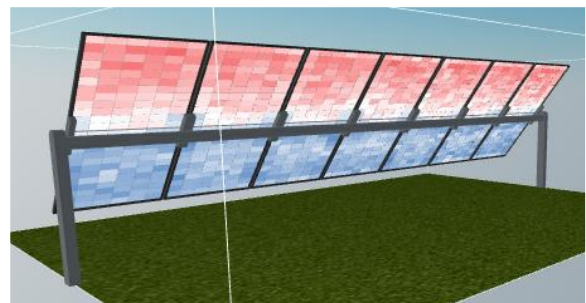


Fig. 1. Example output from SunSolve-Yield for the repeated bay of a single-axis tracker; redder/bluer cells generate more/less current.

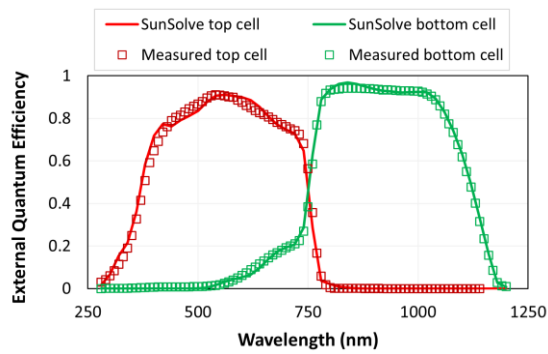


Fig. 2. External quantum efficiency for (red) the top cell and (green) bottom cell of the 2T tandem cell. Data shown as open symbols are measurements reported in [16], while lines represent simulations performed using SunSolve with the optical model applied in this work.

SunSolve power or yield? If SSY, was an IQE used to generate the EQE?

intervals. The current generation is computed in each cell, and then the electrical behavior of the string of modules is solved with SPICE. The simulations also solve and account for module temperature. Fig. 1 plots example output of the cell generation at one time step.

A. Solar cell, module and system

The inputs for the 2TT solar cell were based on the hybrid-processed perovskite-silicon tandem solar cell presented in [16], which had a perovskite bandgap of 1.64 eV. The refractive index of its materials was taken from the supplementary material in [16] (for 25 °C) and are assumed independent of temperature during the optical solving. The electrical model of the 2TT sub-cells consists of one-diode equivalent circuits, where the inputs were determined using curve fitting to the JV curves in figure S3 of [17]. Temperature dependence of the sub-circuits was modeled by applying the PVSyst temperature model to each circuit independently. The coefficients for temperature (μ_{Pmp} , μ_{Isc} , μ_{Voc}) for each sub-cell were based on the values presented in [18]. Whilst this temperature model replicates the trends (and fits the temperature coefficients), future work will develop a more comprehensive and validated circuit and temperature model. A sensitivity analysis (not presented here) revealed significant alterations to the temperature model had minor impact on the simulated P and did not alter the trends presented in this work.

Appendix A summarizes the 2TT layer thicknesses and circuit parameters, and Fig. 2 illustrates the close agreement between simulated and measured external quantum efficiency, providing some confidence in the optical and electrical inputs to the 2TT at 25 °C. The inputs for the PERC solar cell were those used in [12].

The dimensions and materials of the modules (cell dimensions, encapsulation, electrode layout, cell stringing, bypass diodes, frame) were the same as those in [12]. We applied the same module inputs for 2TT and PERC to permit a direct comparison between cell technologies, despite 2TT perovskite-silicon bifacial modules not being commercially available (especially 144 half-cut solar cells).

The modules were installed on a bifacial SAT very similar to that published in [15] and applied in [12]. An array was formed by connecting 560 parallel strings of 10 series-connected modules into a SC3000-EV SMA inverter. This constitutes a field of 5600 tandem modules with an inverter loading ratio (i.e. DC/AC ratio) of 1.3. The albedo was 20% and the SAT included backtracking. Since the illumination on the modules is not uniform (e.g., due to row-to-row shading of the diffuse light), the SPICE solutions incorporate current mismatch between cells in the module; in fact, for 2TT modules, this mismatch also includes current mismatch between the sub-cells.

B. External conditions

The external conditions for each site were taken from TMY3 files generated by Solcast [19], which contained the direct and diffuse irradiance, the ambient temperature, wind speed, atmospheric pressure and precipitable water vapor (PWV) at one-hourly intervals throughout a typical year. Ozone concentration and aerosol optical depth (AOD) at 500 nm were taken from NASA's Giovanni satellite data [20].

Clear-sky spectra were determined with SPECTRL2 [21] using the aforementioned atmospheric inputs and the air mass (AM) calculated from the solar zenith. The cloudy-sky spectra were determined by manipulating the clear-sky spectra following the approach of Ernst *et al.* [22], which effectively assumes neutral density scattering and filtering by the clouds. The spectra were scaled to give the DNI and DHI contained in the TMY3 files.

C. Outputs

We determined f_λ at every time step, as well as an annual f_λ , by comparing solutions for output power that were computed with synthesized spectra to those computed with the AM1.5g spectrum (for both direct and diffuse light). Like several studies [e.g., 13, 23–27], we can then evaluate how f_λ trends with atmospheric conditions. We also determined the bifacial gain by comparing solutions in which real illumination is either prevented or permitted to enter the rear of the module.

III. RESULTS

A. Spectral correction

Fig. 3 plots f_λ for the 2TT and PERC example modules at two locations: Golden, USA and Beijing, China. We selected these locations because they represent sites that have a low and high seasonal variation. Golden has low aerosols and cloud coverage, and the seasonal changes due to AM are partially mitigated by the seasonal changes due to PWV. By contrast, Beijing has much higher aerosols and cloud coverage, which have the impact of compounding the seasonal changes.

Fig. 3 illustrates how both the seasonal impact on f_λ and the scatter in f_λ is much higher in Beijing than in Golden, and much

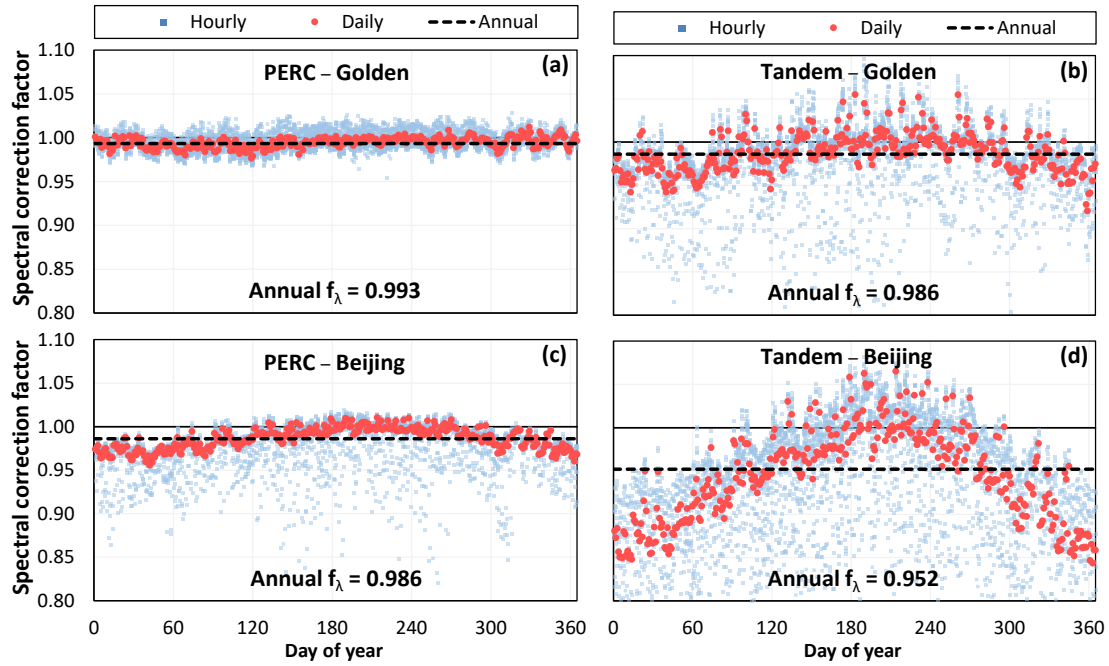


Fig. 3. Spectral correction factor f_{λ} for (left) PERC and (right) 2TT modules installed in an SAT at (top) Golden and (bottom) Beijing.

higher for 2TT than PERC. Table I quantifies these changes. It shows that the seasonal variation in f_{λ} is about 3 times higher—and the scatter is about 4 times higher—for 2TT than PERC.

Another important metric is the annual f_{λ} . It deviates significantly from unity in all cases but is significantly lower for 2TT than PERC at each site. Since these (and practically all) modules are optimized for STC, it should be possible to reoptimize them such that the annual f_{λ} is unity, but that would not necessarily mean that the modules were optimized to give the maximum EY at that particular location and system configuration [28]. While this annual f_{λ} is very important for forecasting EY, more pertinent to this study is the variability in the module performance (i.e., its efficiency) over the course of a day and year.

TABLE I

SIMULATION RESULTS.

SCATTER REPRESENTS THE 95% CONFIDENCE INTERVAL.

Module tech	Location	Annual f_{λ}	Seasonal variation	Scatter, all data
PERC	Golden	0.993	2%	$\pm 2.2\%$
PERC	Beijing	0.986	4%	$\pm 5.4\%$
Tandem	Golden	0.986	5%	$\pm 8.8\%$
Tandem	Beijing	0.952	15%	$\pm 19.4\%$

It is useful to observe the dependence of f_{λ} on atmospheric conditions. Fig. 4 plots f_{λ} against AM, PWV, AOD and O_3 for the Beijing site. Once again, f_{λ} varies much more for 2TT than PERC, but both demonstrate the same trends, whereby f_{λ}

increases as AM and O_3 decrease and PWV and AOD increase. We discussed the reasons for these trends for PERC cells in a previous article [12].

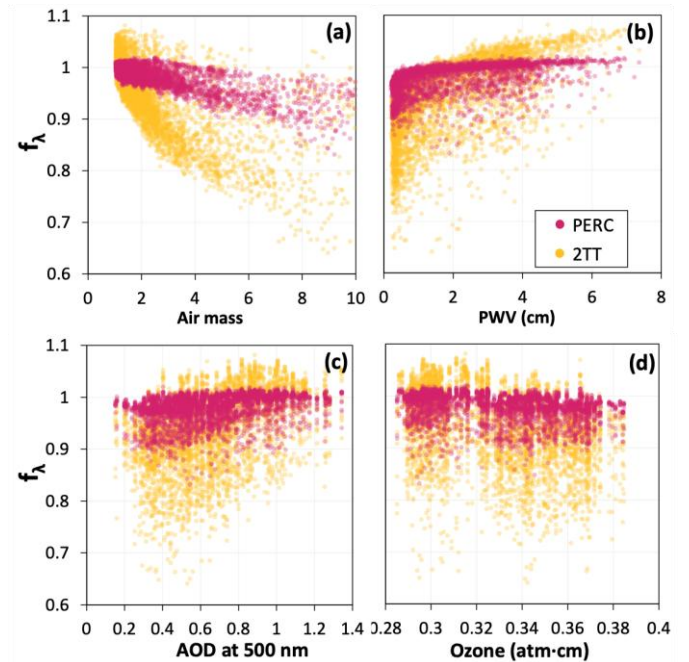


Fig. 4. f_{λ} for bifacial modules simulated at Beijing plotted against (a) air mass, (b) PWV, (c) AOD and (d) ozone for 2T tandem (orange) and PERC modules (red).

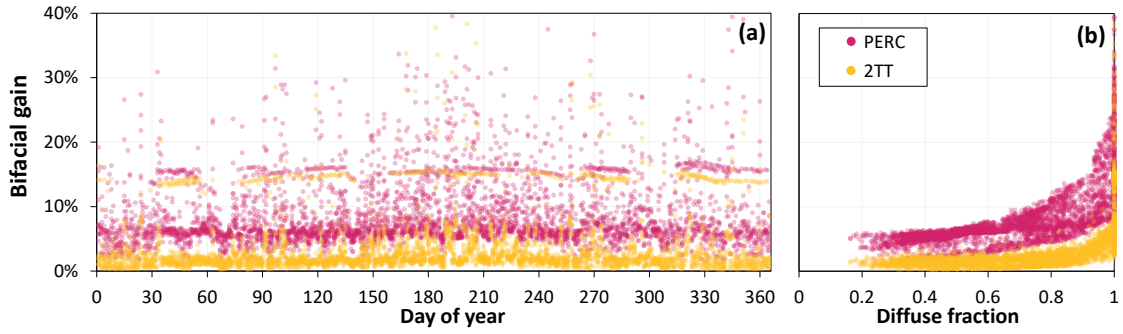


Fig. 5. Bifacial gain of PERC and tandem modules at every time step vs (a) day of year and (b) diffuse fraction, as calculated for Beijing.

B. Bifacial gain

The annual bifacial gain for Beijing was 1.9% and 6.7% for 2TT and PERC modules, respectively. Consistent with [5], the bifacial gain is much smaller for 2TT modules because rear-side illumination is absorbed entirely by the bottom Si cell, and hence there is only a significant boost in power when the 2TT is current-limited by the bottom sub-cell. Thus, optimizing a 2TT cell will differ for bifacial and monofacial applications [5].

Fig. 5 plots the bifacial gain calculated for PERC and tandem modules at each time step vs (a) day of year and (b) diffuse fraction. It shows that the modules exhibit only slight seasonal dependencies; in fact, the seasonal dependence of the 2TT module is greater, showing a slightly higher bifacial gain in the summer than in the winter. This arises because the 2TT module becomes current-limited more often by the Si sub-cell in summer.

Fig. 5(b) shows that the bifacial gain is strongest on cloudy days, when a greater fraction of light reflects from the ground. This effect is more pronounced for PERC cells, which do not suffer from sub-cell current mismatch.

A final observation from Fig. 5 is that bifacial gain varies mostly between 5% and 20% for the PERC cell, with a weighted average of 7%. Since, for c-Si cells, this boost is roughly linearly related to the boost in rear-side irradiance, this

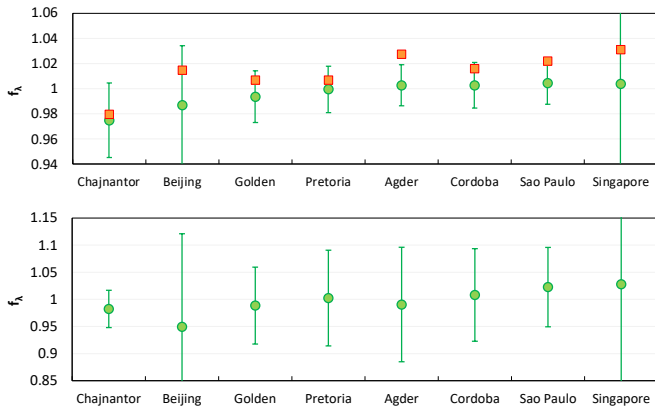


Fig. 6. f_λ at various sites for (a) PERC and (b) 2TT. Green symbols plot annual f_λ , error bars plot the range of hourly f_λ (95% CI), and red symbols plot annual f_λ after applying a cloud correction devised from FARMS-NIT.

illustrates by just how much rear irradiance varies in a tracking system. It demonstrates the impossibility of optimizing a bifacial 2TT for all conditions, i.e., the cells will operate under suboptimal conditions at almost all times, which may have implications for degradation [4].

C. Additional sites

Fig. 6 plots f_λ of the same bifacial SAT at eight sites around the world. The sites were selected to provide a wide variety of cloud, AOD, PWV and latitude. The green symbols plot the annual f_λ , and the error bars represent the 95% confidence interval of the hourly f_λ . These are plotted for (a) PERC and (b) 2TT modules. We note that the variability in f_λ is higher at cloudier sites.

As mentioned above, these simulations computed the cloudy-sky spectra applying the approach by Ernst *et al.* [22], which is easily implemented and quick to solve. A less accessible but significantly more accurate model is contained in FARMS-NIT [29], which accounts for the wavelength-dependent scattering and absorption of clouds.

D. Inverter clipping

Many modern systems are oversized relative to their inverter. This leads to “clipping”, whereby the inverter limits the input power to some maximum, which then forces the module to operate at voltages above V_{MP} . (In addition, systems are also often curtailed by the grid, which also leads to module voltages above V_{MP} .)

These suboptimal operating conditions have implications for 2TT modules: Firstly, it lessens the need to optimize the perovskite layer to maximize P under real conditions. Fig. 7 illustrates this by plotting P into (DC) and out of (AC) the inverter on three summer days at Cordoba, Argentina. It shows how P is clipped on sunny days but not on cloudy days, and that during clipping, f_λ is effectively unity because the system generates the same power whether the modules are submitted to the real or AM1.5g spectrum. Thus, there is less value in optimizing the perovskite layer to maximize P_{AC} for external conditions when the 2TT modules are installed in oversized or

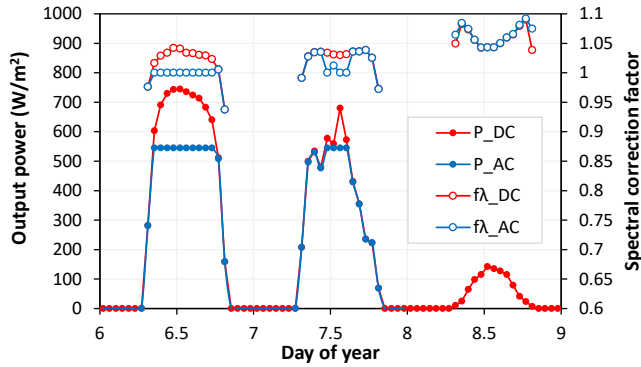


Fig. 7. Output power (solid symbols) and spectral correction (open symbols) for three days of the year at Cordoba, Argentina. Data labeled DC and AC show the power in and out of the inverter.

curtailed systems. Or, at least, the most value would come from optimizing the cells for spectra that arise when the sun is lower in the sky (higher AM).

Secondly, when systems are clipped or curtailed, module voltages operate above V_{MP} , which may have implications for degradation [3]. In short, designing 2TT cells for the real world is complicated.

IV. PARAMETERIZATION FOR EY SOFTWARE

Several major EY software packages, including PVSyst [30], solve P_{DC} at AM1.5g and then provide the user with the option to modify the power at each time step using the First Solar equation (Eq 1 [26]). This model determines f_{λ} using a parameterization with AM and PWV, so it cannot capture all of the spectral dependencies (e.g., the dependencies on cloud cover, aerosols, ozone, pressure and far-field albedo). Nevertheless, it still offers a more accurate approach to accounting for spectrum than assuming $f_{\lambda} = 1$, provided the coefficients of the First Solar equation are calculated for the module, location and system configuration of interest.

We demonstrate the improvement with Fig. 8, which gives simulation results for the 2TT module at Beijing. (Recall that

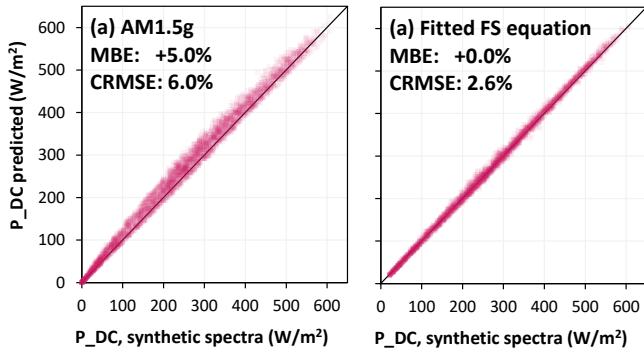


Fig. 8. P_{DC} when assuming the incident spectra is (a) AM1.5g and (b) fitted First Solar equation vs P_{DC} when applying synthetic spectra. Simulations for 2TT module on an SAT at Beijing.

Beijing has high cloud cover and aerosol concentrations, so it provides a stringent test on the First Solar model.)

Fig. 8 plots P_{DC} calculated with the AM1.5g against P_{DC} calculated with synthetic spectra. The results of Fig. 8(a) were not corrected ($f_{\lambda} = 1$), and the results of Fig. 8(b) were corrected by f_{λ} determined from the First Solar equation, where the coefficients were determined by a least-square fit to the results when using the synthetic spectra. Thus, any points on the one-to-one lines of Fig. 8 contain no error in f_{λ} relative to the synthetic spectra.

We find that when spectral effects are neglected, P_{DC} is overestimated at most time steps for the 2TT module at Beijing: the annual overestimate, equivalent to the mean-bias error (MBE), is 5.0%, and the scatter, the centered root-mean-square error (CRMSE), is 6.0%. When correcting for spectrum with the First Solar equation with fitted coefficients, the MBE reduces to zero and the CRMSE halves. Thus, the spectral behavior of EY software like PVSyst can be partially accounted for with a fitted First Solar equation. It should be noted however that this result has compared the clear-sky First Solar model to results synthesized with a simplified cloud model (i.e. the Ernst model). The application of more complex models for spectral synthesis that better capture the impact of clouds (i.e. Farms-NIT) may change this finding for cloudy conditions.

V. CONCLUSION

“Full-stack” simulation and analyses—such as that presented in this study—connects the fields of cell, module and system research. This is particularly valuable for 2TT technologies, much more so than c-Si technologies, because of their sensitivity to spectrum. Full-stack simulation permits cell and module technologists to design and test devices for the ultimate operating conditions, and it assists system engineers in reconfiguring system components for the modules of the future.

VI. ACKNOWLEDGEMENT

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APPENDIX A: INPUT TO SUNSOLVE MODEL

Inputs for the layer thicknesses and circuit inputs are shown in the tables below.

TABLE II: PEROVSKITE-SILICON TANDEM

Layer thicknesses	
MgF ₂	110 nm
IZO	106 nm
SnO ₂	20 nm
C60	19 nm
Perovskite (1.64 eV)	450 nm
Spiro-TTB	12 nm
i-n-ncn-ncp	39 nm
Si	182 μm
a-Si(i)	7 nm
a-Si(p)	8 nm
ITO	75 nm
Electrical details	
<i>Top cell:</i>	
J_{01}	1.31×10^{-14} A/cm ²
m_1	1.3
R_{sh}	$2.04 \text{ k}\Omega \cdot \text{cm}^2$
R_s	$1.22 \Omega \cdot \text{cm}^2$
T coefficient for I_L	0.13 %/°C
T coefficient for m_1	-0.057 %/°C
Band gap for T model E_g	1.12 eV
<i>Bottom cell:</i>	
J_{01}	1.5×10^{-14} A/cm ²
m_1	1.7
R_{sh}	$2.56 \text{ k}\Omega \cdot \text{cm}^2$
R_s	$0.69 \Omega \cdot \text{cm}^2$
T coefficient for I_L	-0.06 %/°C
T coefficient for m_1	0.057 %/°C
Band gap for T model E_g	1.64 eV

REFERENCES

- [1] M. R. Vogt *et al.*, "Introducing a comprehensive physics-based modelling framework for tandem and other PV systems," *Solar Energy Materials and Solar Cells*, vol. 247, Oct. 2022, doi: 10.1016/j.solmat.2022.111944.
- [2] Y. Blom, M. R. Vogt, C. M. Ruiz Tobon, R. Santbergen, M. Zeman, and O. Isabella, "Energy Loss Analysis of Two-Terminal Tandem PV Systems under Realistic Operating Conditions—Revealing the Importance of Fill Factor Gains," *Solar RRL*, vol. 7, no. 8, Apr. 2023, doi: 10.1002/solr.202200579.
- [3] Š. Tomšič, M. Jošt, K. Brecl, M. Topič, and B. Lipovšek, "Energy Yield Modeling for Optimization and Analysis of Perovskite-Silicon Tandem Solar Cells Under Realistic Outdoor Conditions," *Advanced Theory and Simulations*, vol. 6, no. 4, Apr. 2023, doi: 10.1002/adts.202200931.
- [4] S. Orooji and U. W. Paetzold, "Energy Yield Modeling of Perovskite-Silicon Tandem Photovoltaics: Degradation and Total Lifetime Energy Yield," *Energy Technology*, Nov. 2024, doi: 10.1002/ente.202400998.
- [5] P. Tillmann, K. Jäger, A. Karsenti, L. Kreinin, and C. Becker, "Model-Chain Validation for Estimating the Energy Yield of Bifacial Perovskite/Silicon Tandem Solar Cells," *Solar RRL*, vol. 6, no. 9, Sep. 2022, doi: 10.1002/solr.202200079.
- [6] E. Kringlebotten, "Comparing Measured to Modelled Spectral Irradiance for EY Assessments of Tandem PV Modules,"
- [7] R. Schmager, M. Langenhorst, J. Lehr, U. Lemmer, B. S. Richards, and U. W. Paetzold, "Methodology of energy yield modelling of perovskite-based multi-junction photovoltaics," *Optics Express*, vol. 27, no. 8, p. A507, Apr. 2019, doi: 10.1364/oe.27.00a507.
- [8] Y. Gao *et al.*, "Performance optimization of monolithic all-perovskite tandem solar cells under standard and real-world solar spectra," *Joule*, vol. 6, no. 8, pp. 1944–1963, Aug. 2022, doi: 10.1016/j.joule.2022.06.027.
- [9] I. Mathews, S. Lei, and R. Frizzell, "Predicted annual energy yield of III-V/c-Si tandem solar cells: modelling the effect of changing spectrum on current-matching," *Optics Express*, vol. 28, no. 6, p. 7829, Mar. 2020, doi: 10.1364/oe.384620.
- [10] O. Dupré, B. Niesen, S. de Wolf, and C. Ballif, "Field Performance versus Standard Test Condition Efficiency of Tandem Solar Cells and the Singular Case of Perovskites/Silicon Devices," *Journal of Physical Chemistry Letters*, vol. 9, no. 2, pp. 446–458, Jan. 2018, doi: 10.1021/acs.jpclett.7b02277.
- [11] N. Tucher *et al.*, "Energy yield analysis of textured perovskite silicon tandem solar cells and modules," *Optics Express*, vol. 27, no. 20, p. A1419, Sep. 2019, doi: 10.1364/oe.27.0a1419.
- [12] McIntosh, Keith R., Malcolm D. Abbott, and Ben A. Sudbury. "Error in the Yield Forecast of c-Si Systems Due to Spectral Effects." In *2024 IEEE 52nd Photovoltaic Specialist Conference (PVSC)*, pp. 1279-1283. IEEE, 2024.
- [13] G.S. Kinsey, N.C. Riedel-Lyngskær, A.A. Miguel, M. Boyd, M. Braga, C. Shou, R.R. Cordero, B.C. Duck, C.J. Fell, S. Feron, and G.E. Georgiou, 2022. "Impact of measured spectrum variation on solar photovoltaic efficiencies worldwide". *Renewable Energy*, 196, pp.995-1016.
- [14] <https://pvlighthouse.com.au/sunsolve>
- [15] K.R. McIntosh, M.D. Abbott, B.A. Sudbury, and J. Meydbray, 2019. Mismatch loss in bifacial modules due to nonuniform illumination in 1-D tracking systems. *IEEE Journal of Photovoltaics*, 9(6), pp.1504-1512.
- [16] L. Xu, et al., "Accurate Optical Modeling of Monolithic Perovskite/Silicon Tandem Solar Cells and Modules on Textured Silicon Substrates." *PRX Energy* 1, 023005, 2022.
- [17] Z. Xu, H. Bristow, M. Babics, B. Vishal, E. Aydin, R. Azmi, E. Ugur et al. "Reverse-bias resilience of monolithic perovskite/silicon tandem solar cells." *Joule* 7, no. 9 (2023): 1992-2002.
- [18] M. Babics, H. Bristow, A.R. Pininti, T.G. Allen, and S. De Wolf. "Temperature coefficients of perovskite/silicon tandem solar cells." *ACS Energy Letters* 8, no. 7 (2023): 3013-3015.
- [19] <https://solcast.com/>
- [20] <https://giovanni.gsfc.nasa.gov/giovanni/>
- [21] R.E. Bird, and C. Riordan, 1986. Simple solar spectral model for direct and diffuse irradiance on horizontal and tilted planes at the earth's surface for cloudless atmospheres. *Journal of Applied Meteorology and Climatology*, 25(1), pp.87-97.
- [22] M. Ernst, H. Holst, M. Winter, and P.P. Altermatt, 2016. SunCalculator: A program to calculate the angular and spectral distribution of direct and diffuse solar radiation. *Solar Energy Materials and Solar Cells*, 157, pp.913-922.
- [23] W.M. Keogh and A.W. Blakers, 2004. Accurate measurement, using natural sunlight, of silicon solar cells. *Progress in Photovoltaics: Research and Applications*, 12(1), pp.1-19.

- [24] J.A. Caballero, E.F. Fernández, M. Theristis, F. Almonacid, and G. Nofuentes, 2018. Spectral corrections based on air mass, aerosol optical depth, and precipitable water for PV performance modeling. *IEEE Journal of Photovoltaics*, 8(2), pp.552-558.
- [25] D. Dirnberger, G. Blackburn, B. Müller and C. Reise, 2015. On the impact of solar spectral irradiance on the yield of different PV technologies. *Solar Energy Materials and Solar Cells*, 132, pp.431-442.
- [26] M. Lee, and A. Panchula, 2016, June. Spectral correction for photovoltaic module performance based on air mass and precipitable water. In 2016 IEEE 43rd Photovoltaic Specialists Conference (PVSC) (pp. 1351-1356). IEEE.
- [27] M. Schweiger, and W. Herrmann, 2017, June. Characterizing the impact of solar spectral irradiance on pv module output. In 2017 IEEE 44th Photovoltaic Specialist Conference (PVSC) (pp. 1-3). IEEE.
- [28] J. M. Ripalda, D. Chemisana, J. M. Llorens, and I. García, "Location-Specific Spectral and Thermal Effects in Tracking and Fixed Tilt Photovoltaic Systems," *iScience*, vol. 23, no. 10, Oct. 2020, doi: 10.1016/j.isci.2020.101634.
- [29] Y. Xie, M. Sengupta, and C. Wang, 2019. A fast all-sky radiation model for solar applications with narrowband irradiances on tilted surfaces (FARMS-NIT): Part II. The cloudy-sky model. *Solar Energy*, 188, pp.799-812.
- [30] PVSyst, <https://www.pvsyst.com/>